

Evaluating Step-by-step Reasoning Traces: A Survey

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Abstract

Step-by-step reasoning is widely used to enhance the reasoning ability of large language models (LLMs) in complex problems. Evaluating the quality of reasoning traces is crucial for understanding and improving LLM reasoning. However, existing evaluation practices are highly inconsistent, resulting in fragmented progress across evaluator design and benchmark development. To address this gap, this survey provides a comprehensive overview of step-by-step reasoning evaluation, proposing a taxonomy of evaluation criteria with four top-level categories (factuality, validity, coherence, and utility). Based on the taxonomy, we review different evaluator implementations and recent findings, leading to promising directions for future research.

1 Introduction

Large language models (LLMs) have demonstrated remarkable capabilities in reasoning on complex problems, such as logic, math, and science. At the core of this versatility lies **step-by-step reasoning** (Wei et al., 2022; Kojima et al., 2022), where the LLM generates an intermediate reasoning trace before presenting the final answer.

The reasoning ability of LLMs is often measured in terms of *answer accuracy*, i.e., finding the correct answer for a complex reasoning problem (Cobbe et al., 2021; Zhong et al., 2021). However, answer accuracy is generally insufficient for measuring LLMs' reasoning ability, as the correct answer does not imply the correctness of the preceding reasoning trace (Lanham et al., 2023; Mirzadeh et al., 2024; Paul et al., 2024). Furthermore, assessing the quality of the reasoning trace can directly lead to better reasoning ability of LLMs by verifier-guided search (Wang et al., 2023b; Yao et al., 2023; Hao et al., 2024) and reinforcement learning (Lu et al., 2024; Cui et al., 2025).

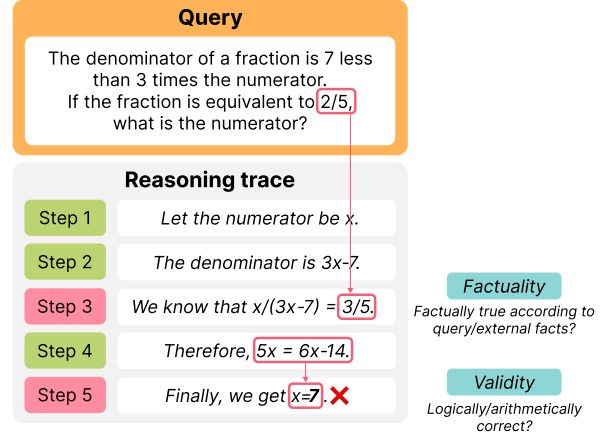


Figure 1: Illustrative example of reasoning trace evaluation.

Consequently, reasoning trace evaluation is an active research topic, with numerous new evaluators and datasets continuously being proposed. However, this rapid growth has led to a proliferation of evaluators and datasets without establishing a consensus on the criteria (what to evaluate). In this survey, we aim to provide a systematic review of existing step-by-step reasoning evaluation criteria, which will serve as a foundation for implementing evaluators.

Implementing an evaluator also introduces several practical decision choices. Different architectures involve trade-offs between computational cost and expected performance, while non-architectural factors like training data and data format also play a significant role. This survey seeks to categorize and compare various evaluator implementations, highlighting key trade-offs and revealing additional dimensions that merit consideration.

The key contributions of this survey are:

- Defining a clear, universal taxonomy of step-by-step evaluation criteria (§3).
- Surveying existing evaluators for step-by-step reasoning evaluation based on their im-

plementations, across diverse reasoning tasks and criteria (§4).

- Identifying recent findings and promising directions for reasoning trace evaluation (§5-§6).

2 Background

2.1 Step-by-step reasoning

Step-by-step reasoning is where LLMs generate a series of intermediate natural language steps ("thoughts") before outputting the final answer (Wei et al., 2022). Each instance consists of two parts: a **query** and a **reasoning trace**, and the **final answer**. Upon seeing the query (user input), the LLM autoregressively generates its solution as a reasoning trace. Finally, a trace should include a **final answer** for the query, which can be compared to the ground truth. See Appendix A for details on different reasoning tasks.

2.2 Evaluation

Reasoning trace **evaluators** assess the quality of the reasoning trace and assign a score, reflecting whether it is good or not based on the criterion. Evaluators can be intrinsic metrics like uncertainty to models specialized for reasoning trace evaluation; see Section 4 for different types of evaluators.

2.3 Meta-evaluation

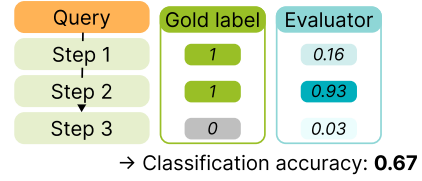
How can we evaluate these evaluators (meta-evaluation)? Two common directions apply: (1) using meta-evaluation benchmarks with step-wise labels, or (2) measuring the improvement in the downstream task performance (Figure 2).

2.3.1 Meta-evaluation Benchmarks

Meta-evaluation benchmarks contains labels indicating a step’s quality based on the predefined criteria. In this setting, the evaluator’s performance is measured by the classification accuracy of these labels. These benchmarks offer a fine-grained view of *which criteria the evaluator can handle well and which cannot* (Song et al., 2025). *However, constructing these data often requires costly human annotation* (Lightman et al., 2024; Zheng et al., 2024a) and the gains in meta-evaluation benchmark *might not generalize to downstream performance* (Zhang et al., 2025). Further details can be found in Appendix B.

Meta-evaluation benchmarks:

Can evaluator classify good/bad steps?



Verifier-guided search:

Can the evaluator choose the most promising trace?



Reinforcement learning:

Are evaluator scores a good reward function?

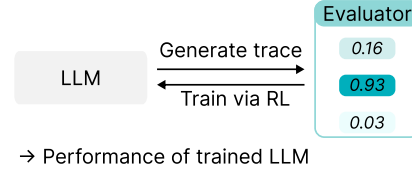


Figure 2: Illustration of three popular meta-evaluation methods: meta-evaluation benchmarks, verifier-guided search, and reinforcement learning.

2.3.2 Downstream performance improvement

As the fundamental goal of evaluators is to improve the reasoning ability of LLMs, the evaluator’s quality can also be measured by the improvement in downstream reasoning tasks.

Verifier-guided search uses evaluator scores to find the most promising trace after exploring different paths. Popular methods include **Best-of-N decoding** (independently sampling N traces and selecting one) (Lightman et al., 2024; Zhang et al., 2024c) and **tree search** (sampling multiple candidate steps and choosing the most promising path) (Yao et al., 2023; Guan et al., 2024; Zhu et al., 2024b). *The performance is often compared to majority voting without evaluators (Self-consistency; Wang et al. (2023b)), where a bigger gap indicates a better evaluator performance.*

Reinforcement learning (RL) uses evaluator scores as a *reward* to further train an LLM (Uesato et al., 2022; Pan et al., 2023b; Zhang et al., 2024a). *If the evaluator provides useful training rewards, the trained model will reach higher final answer accuracy.* Moreover, as evaluators that are vulnerable to spurious features like length lead to *reward hacking*, successful RL also indicates the evaluator’s robustness (Zhang et al., 2024a).

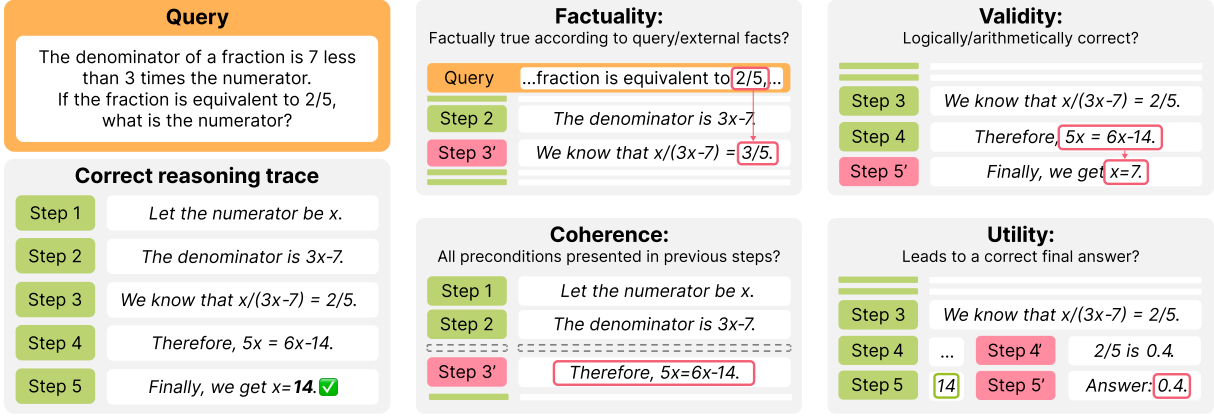


Figure 3: Illustration of the proposed categories of step-by-step reasoning evaluation criteria, *i.e.* factuality, validity, coherence, and utility. The left shows an example of a query and a reasoning trace. The other four blocks demonstrate examples that fail to satisfy the respective metric. Red filled rectangles indicate the error’s location, and the outlined boxes and arrows show the cause of the error. The trace example is originally from Lightman et al. (2024).

3 Evaluation Criteria

Previous studies have proposed various criteria for evaluating step-by-step reasoning (Golovneva et al., 2023a; Lightman et al., 2024; Wang et al., 2024c; Jacovi et al., 2024), but these works failed to propose a complete taxonomy that covers diverse reasoning tasks (*e.g.*, literature in factual reasoning and math reasoning have focused on different criteria). In this section, we propose a unified taxonomy of reasoning trace evaluation criteria that spans different reasoning tasks and evaluators. We categorize them into four key dimensions: **Factuality**, **Validity**, **Coherence**, and **Utility** (Figure 3)¹.

3.1 Factuality

Factuality evaluates if the factual information can be grounded in reliable sources.

The narrower notion of factuality is **groundedness**, where the generated trace should be true according to the *query* (Lewis et al., 2020; Gao et al., 2024d). For instance, if the retrieved document explicitly mentions that *Einstein was born in 1879*, the step mentioning that he was born in *1789* is ungrounded. In less factual tasks like math, groundedness also indicates using correct numbers and constraints given in the query.

However, the reasoning process might require factual knowledge not directly mentioned in the query. This type of factuality can be referred to as **parametric knowledge**. While steps containing trivia-style facts can be readily verified by retrieval-based fact checkers (Thorne et al., 2018), verify-

ing subtle, commonsensical knowledge remains an open challenge (Toroghi et al., 2024).

3.2 Validity

Validity evaluates if a reasoning step contains no logical errors.

The validity of a reasoning step can be defined in terms of *entailment* (Bowman et al., 2015), which is widely accepted in factual/commonsense-based reasoning (Prasad et al., 2023; Wu et al., 2024a). Under this definition, a step is considered valid if it can be directly entailed from previous steps (Tafjord et al., 2021; Dalvi et al., 2021; Saparov and He, 2023) or at least does not contradict them (Golovneva et al., 2023a; Prasad et al., 2023; Zhu et al., 2024b).

In tasks like math or logic, the more common form of validity is *correctness*, *e.g.* performing accurate calculations in arithmetic reasoning (Lightman et al., 2024; Jacovi et al., 2024; Zheng et al., 2024a) or inferring the correct logical conclusion based on the provided premises (Wu et al., 2024b; Song et al., 2025).

3.3 Coherence

Coherence measures if a reasoning step’s *preconditions* are satisfied by the previous steps (Wang et al., 2023a; Lee and Hwang, 2025). For instance, if a trace includes the reasoning step “Next, we add 42 to 16.” but the origin of the value 42 was never explained in the previous steps, this step is considered incoherent. An intuitive way to obtain an incoherent trace is randomly shuffling a coherent trace (Wang et al., 2023a; Nguyen et al., 2024), as

¹These criteria are independent but not mutually exclusive (a step can fail to satisfy multiple criteria).

the premise of some steps will not appear anywhere in the previous steps (*incoherent*) even though it can be eventually deduced from the query (*valid*).

Note that coherence judgment is inherently subjective and pragmatic compared to other criteria (Jacovi et al., 2024). For instance, seemingly trivial steps like "A part of something is present in that something" in WorldTree V2 (Xie et al., 2020) is annotated as necessary in Dalvi et al. (2021) but not necessary in Ott et al. (2023).

3.4 Utility

Utility measures whether a reasoning step contributes to getting the correct final answer.

The narrower interpretation of utility is *progress*, or whether the step is correctly following the ground truth solution (Saparov and He, 2023; Nguyen et al., 2024). For instance, in Game of 24 (making the number 24 using 4 natural numbers and basic arithmetic operations) (Yao et al., 2023), a solution can be defined as a sequence of operations (e.g. $5 + 7 = 12 \rightarrow 12 - 6 = 6 \rightarrow 6 \times 4 = 24$). In this task, the utility of a step (making $5 + 7 = 12$ from 5 and 7) can be directly assessed by checking if it is a part of a correct solution.

The more general version of utility is *value function* (estimated reward). (Chen et al., 2023; Wang et al., 2024c; Setlur et al., 2024). Value function is often measured using Monte Carlo Tree Search (MCTS), where the step’s value is determined by the average/maximum reward of sampled continuations. Evaluating utility as a value function offers high scalability as it only requires the gold answer for computing the reward, without any human annotation or ground-truth solutions (Wang et al., 2024c; Lai et al., 2024; Cui et al., 2025).

4 Evaluator types

The goal of reasoning trace evaluators is to assess reasoning traces by assigning scores. However, choosing the right evaluator for the target criteria and task is non-trivial. For instance, there is no guarantee that evaluators designed for factuality and multi-hop question answering will seamlessly work on math reasoning problems.

In this survey, we provide a comprehensive overview of diverse reasoning trace evaluators, (Luo et al., 2024a; Wei et al., 2025). We summarize eight popular evaluator types based on the criteria they evaluate (summarized in Table 1), along with other practical strengths and weaknesses.

Metric impl.	F	V	C	U
Rule-based	▲	▲	▲	▲
Uncertainty	●		▲	
$\mathcal{V}$ -information		▲	▲	●
LLM-as-value-function				●
Cross-encoder	●	●	▲	▲
Sequence classifiers	▲	●		●
Critic models	●	●	▲	●
Generative verifiers		▲		

Table 1: Mapping between each metric implementation type to the category commonly used, where the acronym FVCU corresponds to factuality, validity, coherence, and utility, respectively. For each combination of metric and implementation, ● denotes that there are at least 3 published works, and ▲ denotes that there are 1 or 2. The full table can be found in Table 2.

4.1 Rule-based matching

For tasks where the ground truth solution can be expressed as a *graph of entities*, a step corresponds to a directed edge between two entities, as in knowledge graphs for factual reasoning (Nguyen et al., 2024) or computation graphs for arithmetic problems (Li et al., 2023b). In this setting, factuality reduces to identifying the correct relation between entities, coherence to the correct ordering of steps, and utility to the existence of the step in the gold reasoning chain (Nguyen et al., 2024; Saparov and He, 2023). However, this approach does not generalize for tasks without clear symbolic representations, e.g., commonsense reasoning or complex math reasoning beyond arithmetic word problems.

4.2 Intrinsic metrics

Uncertainty Uncertainty of the model can be used as an intrinsic proxy about the generated content’s quality (Xiao and Wang, 2021; Zhang et al., 2023b). Qiu et al. (2024) use *token probability entropy*, defined as $\sum_{t \in V} p(t) \log(p(t))$ where p is the probability distribution of all tokens in vocabulary V . Farquhar et al. (2024) and Kossen et al. (2024) extend the approach by clustering semantically similar tokens and calculating the entropy for each cluster. While uncertainty-based evaluators are primarily used for factuality (Wu et al., 2024a; Farquhar et al., 2024), they have also been applied for evaluating validity (Zhu et al., 2025) or utility (Hu et al., 2024), indicating that uncertainty can be a criteria-agnostic proxy of the quality of reasoning steps.

$\mathcal{V}$ -information Chen et al. (2023); Prasad et al. (2023) adopt $\mathcal{V}$ -information (VI) (Hewitt et al.,

Type	Name	Domains	Criteria	Note
Rule-based	DiVeRSe ^E (Li et al., 2023b)	Arith	VU	Computation graph
	Direct Evaluation (Nguyen et al., 2024)	Factual	FVCU	Knowledge graph
Uncertainty	UoT (Hu et al., 2024)	Common, Expert	U	
	Entropy-based decoding (Qiu et al., 2024)	Factual	F	
	Semantic entropy probes (Farquhar et al., 2024; Kossen et al., 2024)	Factual, Common	F	
	SynCheck ^E (Wu et al., 2024a)	Factual	F	
	UnCert-CoT (Zhu et al., 2025)	Code	V	
V-information	REV (Chen et al., 2023)	Common	U	
	ReCEval ^E (Prasad et al., 2023)	Common, Arith	CVU	
	EPVI (Wang et al., 2024d)	Arith, Common	U	
LLM-as-value-function	GenRM (Mahan et al., 2024)	Math, Logic, Code	U	
	V-STaR (Hosseini et al., 2024)	Arith, Code	U	
	MCTS-DPO (Xie et al., 2024)	Math, Common, Science	U	
	Step-DPO (Lai et al., 2024)	Math	U	
	Tree-PLV (He et al., 2024b)	Math, Common	U	
	Step-Controlled DPO (Lu et al., 2024)	Math	U	
	IRPO (Pang et al., 2024)	Math, Common	U	
	PRIME (Cui et al., 2025)	Math	U	
Cross-encoders	ROSCOE-LI (Golovneva et al., 2023a)	Common, Arith	FVC	Off-the-shelf
	ReCEval ^E (Prasad et al., 2023)	Common, Arith	CVU	Off-the-shelf
	DiVeRSe ^E (Li et al., 2023b)	Arith	VU	Off-the-shelf
	DBS (Zhu et al., 2024b)	Common, Arith, Symbolic	FVCU	Synthetic data
	SynCheck ^E (Wu et al., 2024a)	Factual	F	Off-the-shelf
Sequence Classifiers	GSM8k-verifier (Cobbe et al., 2021)	Arith	U	Outcome
	PRM800K (Lightman et al., 2024)	Math	V	Process
	MATH-Minos (Gao et al., 2024a)	Math	V	Outcome/process
	Math-Shepherd (Wang et al., 2024c)	Math	U	Process
	Eurus-PRM (Yuan et al., 2024)	Math	U	Process
	PAV (Setlur et al., 2024)	Math	U	Process
	ReasonEval (Xia et al., 2025)	Math	V	Process
	Qwen-PRM (Zhang et al., 2025)	Math, Science	VU	Process
	VersaPRM (Zeng et al., 2025)	Expert	FV	Process
Critic models	Verify-CoT (Ling et al., 2023)	Math, Symbolic	V	Partial context
	Tree-of-thoughts (Yao et al., 2023)	Arith, Common	U	No fine-tune
	RAGTruth (Niu et al., 2024)	Common	F	
	CPO (Zhang et al., 2024d)	Factual, Arith	U	
	F ² -Verification (Wang et al., 2024b)	Common, Symbolic, Arith	FV	No fine-tune
	OCEAN (Wu et al., 2024c)	Factual, Common	F	
	Critic-CoT (Zheng et al., 2024b)	Math, Common, Science, Logic	U	
	AutoRace (Hao et al., 2024)	Math, Common, Logic	Custom	No fine-tune
	R-PRM (She et al., 2025)	Math	V	
	PARC (Mukherjee et al., 2025)	Math	V	No fine-tune, Partial context
	Reasoning evaluators (Kim et al., 2025b)	Math, Science, Code	V	No fine-tune
	ThinkPRM (Khalifa et al., 2025)	Math, Science, Code	V	
Generative verifiers	CLOud (Ankner et al., 2024)	Math, Logic, Code	V	
	Generative verifier (Zhang et al., 2024c)	Math, Symbolic	V	

Table 2: Evaluators for step-by-step reasoning, grouped by implementation type (^E denotes that the method is an ensemble of different methods). The **Domain** column specifies the domains used for meta-evaluating each evaluator. "Arith" refers to arithmetic reasoning tasks, "Common" denotes commonsense question answering, and "Expert" corresponds to specialized expert domains like biomedical, legal, and financial reasoning (Appendix A). When evaluators are tested on both arithmetic and general math tasks, only "Math" is listed. The acronym FVCU in the **Criteria** column represents factuality, validity, coherence, and utility, respectively. For AutoRace (Hao et al., 2024), LLMs are instructed to list the criteria based on incorrect traces.

2021) from information theory. Informally, VI measures if a model family $\mathcal{V}$ can generate the correct goal string g with higher probability when the target string t is given to the model. Formally, $VI(t \rightarrow g) = \log p_{\mathcal{V}}(g|t) - \log p_{\mathcal{V}}(g|\phi)$ when ϕ is an empty string. When g is the final answer and t is the trace, VI becomes the difference between the answer token’s probability between Chain-of-thought reasoning and zero-shot reasoning, which indicates how much information the trace provides to predicting the final answer (utility) (Chen et al., 2023). When g is a step and t is the list of previous steps, high VI means that a step is likely to follow from the context, which roughly corresponds to coherence (Prasad et al., 2023).

LLM-as-value-function RL can train LLMs to align rewards to token probabilities (relative to the *base probability* obtained from the initial model), with training objectives like DPO (Rafailov et al., 2023) and GRPO (Shao et al., 2024). For instance, when the reward is determined by the final answer correctness, the token probabilities directly correspond to utility (Mahan et al., 2024; Lai et al., 2024; Xie et al., 2024; Pang et al., 2024). Unlike sequence classifiers that lose their trace generation ability after fine-tuning, these models retain (and improve) the ability to generate traces. However, this method requires numerous good and bad reasoning traces for training. Consequently, most existing LLM-as-value-function evaluators focus on utility, as it is easier to scale up the data by simply checking if the final answer is correct.

4.3 External evaluators

Cross-encoders Cross-encoders simultaneously encode multiple sentences using a single, small network typically with millions of parameters (Devlin et al., 2019; Liu et al., 2019). They have been widely applied to solve tasks such as natural language inference (Bowman et al., 2015) and fact verification (Thorne et al., 2018), *where one has to determine if the hypothesis can be inferred from the given premise*. Cross-encoders trained on off-the-shelf tasks (Golovneva et al., 2023a; Zha et al., 2023; Prasad et al., 2023) or LLM-perturbed data (Zhu et al., 2024b) can be used to evaluate a reasoning step based on the query (groundedness) or previous steps (validity). However, their limited language understanding ability and shorter context length restrict their performance in more complex tasks.

Sequence classifiers (Reward Models)² Sequence classifiers are language models with a lightweight classification head attached to the final hidden state, *trained to predict a numeric score in a supervised manner* (Lightman et al., 2024; Wang et al., 2024c; Setlur et al., 2024). Sequence classifiers can be further divided into (1) process (step-level) evaluator vs. outcome (trace-level) evaluator based on the granularity of each step (Lightman et al., 2024), and (2) *validity evaluator* vs. *utility evaluator* based on the source of the training data. These models achieve significant performance and efficiency (Cobbe et al., 2021; Zhang et al., 2025), but they often require costly stepwise labels for training (Lightman et al., 2024). Furthermore, they cannot generate rationales for a high or a low score, having limited explainability and leading to spurious errors (Ankner et al., 2024; She et al., 2025).

Critic models (LLM-as-a-judge) Critic models are LLMs that are trained or prompted to evaluate the reasoning traces (Zheng et al., 2023; Kim et al., 2024a; Zheng et al., 2024a; Lin et al., 2024). This approach views trace evaluation as one of many reasoning tasks, where common techniques like *Chain-of-thought prompting* (Huang et al., 2024a) and *reinforcement learning with verifiable rewards* (Chen et al., 2025a) can apply. Numerous works show that LLMs are versatile critics; they can effectively evaluate factuality, validity, coherence, and utility in diverse reasoning tasks with or without fine-tuning (Yao et al., 2023; Jacovi et al., 2024; Wu et al., 2024d; Niu et al., 2024). While conceptually simple and compatible with closed-source models, *generating the rationales requires significant execution time and computation compared to other evaluator types*.

Generative Verifiers This paradigm lies in the middle ground of sequence classifiers and critic models. These models first autoregressively generate the evaluation rationale as critic models do. When the generation terminates, like sequence classifiers, a small, fine-tuned head predicts the scores conditioned on both the original reasoning trace and evaluation rationales generated by itself (Ankner et al., 2024; Zhang et al., 2024c).

²While *reward model* generally refers to any model that predicts the desirability of an action in reinforcement learning, the term ‘(process/outcome) reward model’ in the context of reasoning trace evaluation often refers to the sequence classifier architecture.

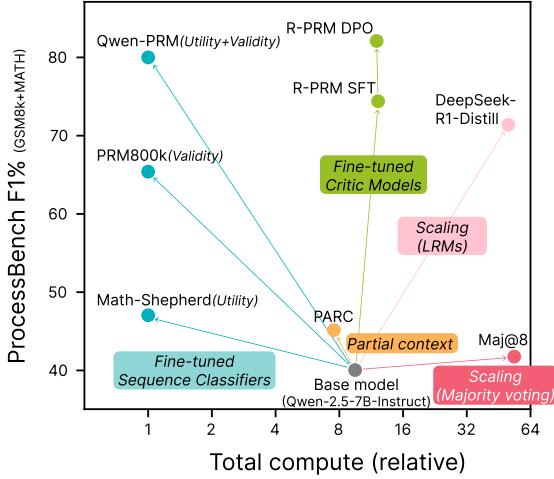


Figure 4: Plot of different evaluators introduced in Section 5, plotted by ProcessBench performance (Zheng et al., 2024a) (GSM8k, MATH subsets averaged) versus total compute for evaluating a trace. While these evaluators share the same base model (Qwen-2.5-7B), they improve the base model’s trace evaluation capability in different ways. Details can be found in Appendix D.

5 Further improving evaluators

This section discusses some of the recent empirical findings on improving evaluators beyond choosing different types, *e.g.*, training data, input format, and scaling compute.

Validity and utility are complementary Validity measures if the step is logically correct, while utility measures if the step makes progress towards the correct answer. Initially, utility-based process reward models were proposed as an *alternative* for validity, since constructing validity data often requires a costly annotation process (Wang et al., 2024c). **Under the hood, there lies an implicit assumption that useful steps are mostly valid.**

However, recent works show that the two criteria are rather *complementary*, from training sequence classifiers to using critic models. Zhang et al. (2025) trains a sequence classifier by only considering steps that are both valid (judged by critic models) and useful (by MCTS-based rollouts) steps as positive, substantially improving performance over baselines that only consider validity or utility (Figure 4, *Sequence Classifiers*). Sun et al. (2024); Kim et al. (2025b) has also shown that averaging validity and utility scores from critic models results in better performance than individual scores in Best-of-N decoding.

The misalignment between validity and utility is

mainly caused by steps that are logically wrong but reach the correct answer (Zheng et al., 2024a; Wang et al., 2025b; Kim et al., 2025b). These invalid but useful steps, also known as *unfaithful reasoning* of LLMs (Lyu et al., 2023; Schnitzler et al., 2024), might lead to overestimation of reasoning ability (Lyu et al., 2023; Petrov et al., 2025).

Partial context allows efficient and accurate evaluation Validity and coherence evaluate a step based on its previous steps. The most intuitive way is to use the *full context* (all preceding steps). However, this approach is not feasible when the trace exceeds the context length of the evaluators, *e.g.*, large reasoning models’ traces are often too long to apply critic models (Kim et al., 2025b).

An alternative solution is to use a *partial context*, where only relevant parts of the query and preceding steps are selected and passed to the evaluator (Ling et al., 2023; Mukherjee et al., 2025). These works first construct a directed *entailment graph*, and evaluate the step only based on the identified premises. This allows evaluators to use shorter context, which is both computationally efficient (Ling et al., 2023) and even more accurate as distractors are removed from the context (Mukherjee et al., 2025) (Figure 4, *Partial context*). Moreover, the graph structure also distinguishes direct errors (premises are valid but the reasoning is invalid) and accumulated errors (premises are invalid but reasoning is valid) (Mukherjee et al., 2025).

Test-time scaling improves evaluator performance Test-time scaling is a general paradigm where investing more test-time compute leads to improved performance. Test-time compute can be scaled in diverse directions, such as sampling the output multiple times (Wang et al., 2023b; Yao et al., 2023) or generating more tokens during a single inference (Snell et al., 2024; Qwen-Team, 2024; DeepSeek-AI, 2025).

This paradigm can be extended to critic models that reason on reasoning traces, especially in meta-evaluation benchmarks. When applying majority voting of independently sampled K evaluator scores in generative models (critic model, generative verifiers), the accuracy in predicting incorrect steps increases linearly with the scale of $\log K$ (Singhi et al., 2025; Kim et al., 2025b; She et al., 2025) (Figure 4, *Majority voting*). Furthermore, using large reasoning models (LRMs) with stronger reasoning capability by generating longer traces (Zheng et al., 2024a; Kim et al., 2025b; Khalifa

et al., 2025) leads to significant improvement in error detection (Figure 4, *Fine-tuned LLMs*).

In verifier-guided search settings, one can either scale exploration or scale evaluation. For instance, in Best-of-N decoding, one can increase the number of responses or use critic models that produce longer outputs. What is the optimal strategy with a constrained computing budget? For relatively weaker evaluators, simple majority voting (Wang et al., 2023b) often outperforms verifier-guided search (Zhang et al., 2025; Singhi et al., 2025). However, using stronger evaluators, e.g., sequence classifiers with better training data (Zhang et al., 2025) or critic models with stronger reasoning capabilities (Khalifa et al., 2025; Kim et al., 2025b) for Best-of-N can effectively outperform majority voting using the same computation budget.

6 Future directions

Evaluating real-world reasoning traces with external knowledge Existing datasets for reasoning trace evaluation are mostly restricted to simple factual reasoning (e.g. factual multi-hop question answering) or self-contained problems (e.g. math problems). However, many realistic reasoning tasks such as repository-level coding (Zhang et al., 2023a), medicine (Savage et al., 2024), and law (Holzenberger and Van Durme, 2021; Kim et al., 2024b) require external up-to-date knowledge retrieval-augmented generation (Lewis et al., 2020). Developing evaluators and meta-evaluation benchmarks for these tasks will significantly enhance the applicability of reasoning trace evaluation in more realistic scenarios.

Evaluating long, complex reasoning traces Following OpenAI o1 (OpenAI, 2024b), numerous large reasoning models (LLMs) that generate long, complex traces involving self-verification and backtracking were introduced (DeepSeek-AI, 2025; Muennighoff et al., 2025; Gandhi et al., 2025). However, existing evaluators are not suitable for these complex traces. For instance, assigning a single scalar score (e.g., sequence classifiers) will make invalid steps corrected afterwards (*Wait, this reasoning is not correct.*) and ones not corrected indistinguishable. Since LLM reasoning traces can contain critical errors (Petrov et al., 2025; Chen et al., 2025b), the effort to develop evaluation resources for such traces will lead to a better understanding of LLMs’ behaviors and further improvement in their performance and credibility.

Symbol-grounded evaluation of reasoning traces

Reasoning tasks often have a symbolic ground truth solution. For instance, deductive reasoning tasks can be represented with formal logic, and arithmetic problems can be expressed as a series of equations or symbolic theorems. These solutions provide precise, formal ways to define evaluators, including validity and utility (progress). However, not much work has been done to exploit the parallel between reasoning traces and the underlying symbolic solution. While several rule-based approaches parse reasoning traces for evaluation in relatively simpler reasoning tasks (Saparov and He, 2023; Nguyen et al., 2024; Li et al., 2023b), no attempts have been made to extend this paradigm to evaluate reasoning traces for more complex and realistic tasks like first-order logic reasoning (Han et al., 2024a,b) and formal math reasoning that use interactive theorem provers (e.g., Lean, Isabelle) (Yang et al., 2023; Gao et al., 2024c).

Objective metrics for coherence evaluation.

LLMs often omit trivial inference steps in their reasoning (Saparov and He, 2023; Prasad et al., 2023), but even human annotators significantly disagree on whether a certain step can be taken as granted (Jacovi et al., 2024). As a result, we do not have many coherence evaluators or datasets (Table 1, 3). A human-annotated dataset of omissible and non-ommissible steps would facilitate the development of practical coherence evaluators and partial context extractors.

7 Conclusion

This survey aims to organize existing criteria and methods for step-by-step reasoning evaluation, which is crucial for understanding and improving LLM’s reasoning capabilities. We provide a unified taxonomy for evaluation criteria, a comprehensive review of existing evaluators and their implementation, and examine recent directions on how to improve these evaluators.

Still, diverse challenges remain in evaluating step-by-step reasoning traces. As new reasoning tasks and methods emerge, existing evaluators often become obsolete for evaluating complex reasoning traces from new tasks and models. As LLMs are now involved in challenging and high-stakes reasoning tasks in the real world, understanding the nature of their errors and precisely evaluating the reasoning trace will remain important.

8 Limitation

References This survey includes an extensive list of recent publications (mostly between 2022 and 2025) on reasoning trace evaluation, sourced from *ACL, EMNLP, NeurIPS, and arXiv preprints, etc. While there might be missing references due to the sheer volume of works produced in this field, we will continue to update missing references and newly released impactful works that contribute to the field.

Survey on diverse empirical results While Figure 4 contains a controlled comparison between different approaches like training sequence classifier with different data, using partial context, or applying test-time scaling techniques, the comparison is limited to ProcessBench results for two reasons: (1) While most paper report reasoning performance improvement results (Section 2.3), these results are often not directly comparable because they make use of different base model, which strongly affect the overall performance. (2) Other meta-evaluation benchmarks than ProcessBench (Jacovi et al., 2024; Zeng et al., 2024a; Song et al., 2025) have not been applied to diverse evaluator implementations at the time of writing.

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A Tasks

This section describes different reasoning tasks and datasets in more detail. While all reasoning tasks fundamentally share the same criteria, literature about a specific task has focused on one criterion over others. For instance, evaluators for factual reasoning tasks often emphasized detecting infactual statements, while evaluators for math reasoning tasks aimed for invalid statements. These discrepancies are one of the fundamental causes of the divergence of the terminologies and definitions in the field.

A.1 Multi-hop Question Answering

Multi-hop question answering (MHQA) tasks require taking information from multiple sources to derive the correct answer (Yang et al., 2018). MHQA is often divided into two subcategories, **factual reasoning** and **commonsense reasoning**.

Answering factual MHQAs can be seen as finding the sequence of *bridging entities* that leads to the final answer (Yang et al., 2018; Talmor and Berant, 2018; Kwiatkowski et al., 2019). For example, to solve a factual MHQA question "*The Argentine PGA Championship record holder has won how many tournaments worldwide?*", one must first find who the Argentine PGA championship record holder is (bridging entity) and determine how many tournaments he has won worldwide. As bridging entity identification does not require sophisticated reasoning ability compared to other tasks, reasoning trace evaluation on factual MHQA mostly focuses on the factuality based on semantic alignment between the query (retrieved documents) and the trace (Golovneva et al., 2023a).

In contrast, an inference step in commonsense MHQAs (Clark et al., 2018; Mihaylov et al., 2018; Talmor et al., 2019; Bisk et al., 2019; Geva et al., 2021; Trivedi et al., 2022) can require information that is not present in the query. The form of such commonsense knowledge can be diverse, ranging from well-known facts (*Paris is in France.*) to logical rules (*If A was born after B was dead, they have never met each other*). Due to these implicit steps, factuality, validity, and coherence are often hard to separate in evaluating commonsense reasoning traces (Jacovi et al., 2024; Zeng et al., 2024b). Furthermore, due to the inherent subjectiveness of validity and coherence in commonsense reasoning, there might be non-negligible inter-annotator disagreement on certain questions (Jacovi et al., 2024).

LLMs are known to achieve strong performance in challenging MHQA datasets such as ARC-Challenge and PIQA, sometimes exceeding human performance (OpenAI, 2024a; Anil et al., 2023). However, multiple studies report that even modern LLMs like GPT-4 (OpenAI, 2024a) are vulnerable to errors, such as failing to correctly adhere to long evidence (Zhu et al., 2024a), leveraging shortcuts (Schnitzler et al., 2024), or ignoring the temporal relation between events (Li et al., 2024a). Therefore, identifying and categorizing mistakes made by LLMs in these tasks is still an important goal.

A.2 Symbolic Reasoning

Since the discovery of Chain-of-thought prompting (Wei et al., 2022; Kojima et al., 2022), step-by-step reasoning largely expanded LLMs’ ability to solve symbolic reasoning tasks³ such as **mathematical reasoning**, **logical reasoning**, and **algorithmic reasoning**. As the final answer and the reasoning process are highly objective in these tasks, utility and validity are the two most popular criteria for evaluating reasoning traces from symbolic tasks.

Arithmetic reasoning, where the model has to predict the correct answer from arithmetic word problems, is the most renowned variant of math reasoning. Popular benchmarks include MathQA (Amini et al., 2019) and GSM8k (Cobbe et al., 2021), which provide long, diverse natural language queries in contrast to relatively synthetic, simple benchmarks (Koncel-Kedziorski et al., 2016; Miao et al., 2020). Game of 24 (Yao et al., 2023) and Mathador (Kurtic et al., 2024) ask to combine given numbers and arithmetic operations to generate the target number, requiring exploration and backtracking in the exponential solution space.

The recent saturation of LLMs in arithmetic word problems facilitated more challenging **mathematical reasoning** benchmarks from math competitions and university textbooks, covering fields like calculus, probability, statistics, geometry, number theory, and more (He et al., 2024a; Gao et al., 2024b; Glazer et al., 2024; Zhang et al., 2024b). While these benchmarks were highly challenging to the state-of-the-art LLMs of the time of release, recently emerging large reasoning models (OpenAI,

2024b; Qwen-Team, 2024; DeepSeek-AI, 2025) achieve unprecedented performance in these benchmarks by generating long reasoning traces often with self-verification and backtracking.

Deductive logical reasoning (Tafjord et al., 2021; Tian et al., 2021; Saparov and He, 2023; Han et al., 2024a) mainly focuses on logical deduction, where one should repeatedly apply the general rules to specific facts as in classical syllogism. **Constraint-based reasoning** (Zhong et al., 2021; Tyagi et al., 2024) is a variant of deductive reasoning where one must find the solution that satisfies the provided initial constraints (*e.g.*, grid puzzles (Zhong et al., 2021)). As these datasets are easy to solve in a symbolic form like logic programming (Saparov and He, 2023; Pan et al., 2023a; Olausson et al., 2023; Lee and Hwang, 2025) but harder in natural language due to the size of the search space (Kang et al., 2024), they have served as a diagnostic benchmark for understanding and analyzing the complex reasoning capability of large language models (Sinha et al., 2019; Saparov and He, 2023; Han et al., 2024a). However, as these datasets are often synthetically generated from their symbolic representations, they might not fully generalize to real-world problems with linguistic diversity and commonsense.

Finally, **algorithmic reasoning** tasks include manipulating strings and data structures, such as concatenating the last letters of the given words (Wei et al., 2022) or completing the incomplete Dyck language. BIG-Bench-Hard (BBH; Suzgun et al. (2022)) and NPHardEval (Fan et al., 2024) include 11 and 9 algorithmic reasoning tasks, respectively, which are challenging for modern LLMs like GPT-4 and PaLM-540B.

A.3 Others

Science reasoning tasks lie between factual/commonsense reasoning tasks and symbolic reasoning tasks, as they often require understanding complicated facts combined with world knowledge and performing precise math/logical reasoning (Hendrycks et al., 2021; Rein et al., 2024; He et al., 2024a; Lu et al., 2025). The most popular benchmark in this field, GPQA-Diamond (Rein et al., 2024), contains 546 questions from physics, chemistry, and biology, where human experts only get 65% of the problems correct.

Expert-domain reasoning includes domain-specific reasoning tasks that often require significant expertise in the field, *e.g.*, biomedical reason-

³While symbolic reasoning may strictly refer to *algorithmic reasoning* in some literature (Wei et al., 2022; Suzgun et al., 2022), we adopt the broader sense including math and logical reasoning that can be readily expressed in symbols (*e.g.*, equation, logic) (Sprague et al., 2024).

ing (Šuster and Daelemans, 2018; Savage et al., 2024; Zuo et al., 2025), legal reasoning (Holzenberger et al., 2020; Guha et al., 2023; Kim et al., 2024b), and financial reasoning (Chen et al., 2022; Li et al., 2024b). These tasks require both domain-specific knowledge and reasoning strategies, posing a significant challenge to modern language models (Zuo et al., 2025; Li et al., 2024b). However, due to the high cost of expert annotation, existing methods often oversimplify real-world challenges (Holzenberger and Van Durme, 2021; Guha et al., 2023); consequently, the demand for resources that closely reflect real-world expert applications is rising.

Programming/coding is closely related to algorithmic reasoning. Popular benchmarks regarding programming include *competitive coding*, where one has to solve an algorithm problem given in natural language and test codes (Chen et al., 2021; Li et al., 2022), and *practical coding* that covers tasks of software engineers and developers (Zhang et al., 2023a; Jimenez et al., 2024; Chan et al., 2024). Programming differs from other reasoning tasks in various aspects: (1) codes are strictly constrained by predefined syntax and semantics, and (2) the result is evaluated by the execution result rather than the code itself. These constraints make (1) segmenting the trace (code) into steps and (2) applying metrics that require explicitly stated answers, *i.e.*, $\mathcal{V}$ -information, difficult than in natural language reasoning traces. Therefore, most evaluators specialized in code focus on trace-level utility rather than step-wise evaluation, defined as the pass rate of predefined unit tests (Dai et al., 2025).

B Meta-evaluation Datasets

Datasets that annotate LLM-generated reasoning traces can be used to train an evaluator or perform meta-evaluation on different metrics. A summary of existing datasets can be found in Table 3. This section mainly focuses on the dataset construction process, namely data annotation and label types.

B.1 Data collection process

B.1.1 Labeling methods

Human annotation The most straightforward approach to decide the ground truth label is to use *human evaluation* (Lightman et al., 2024; Jacovi et al., 2024; Zeng et al., 2024a; Zheng et al., 2024a). The largest human annotation experiment was conducted by Lightman et al. (2024), where crowdsourced annotators labeled the validity of

800k steps (75k reasoning traces). Due to the sheer volume of annotation, an active learning strategy was used; the annotators were requested to annotate *hard* samples (the final answer is incorrect but judged as valid by the reward model), which were added to the training data for the next version of the reward model.

LLM annotation As a cheap alternative for human evaluation, LLM-as-a-judge is often used to generate labels (Gao et al., 2024a; Zhang et al., 2025). However, LLM-assigned labels are not fully credible, given that state-of-the-art LLMs still make errors in human-annotated datasets (Zheng et al., 2024a; Kim et al., 2025b). Therefore, LLM-annotated data is often used to augment the training data rather than for meta-evaluation purposes.

Perturbation Another method to create positive and negative samples is to insert errors into correct reasoning traces. For instance, Zhu et al. (2024b); Lu et al. (2024) samples traces that reach the correct answer, and prompts an LLM to introduce a predefined form of perturbation to the reasoning trace. This allows easy sampling of diverse erroneous traces that can improve the robustness of evaluators, but using human-defined errors might not correctly reflect the true distribution of LLM-generated errors.

Step-level utility Some datasets use step-level utility as their labels. The most prominent approach is *Monte Carlo Tree Search* (Wang et al., 2024c), where the step-level utility is measured by sampling *rollouts* from a step and checking if they reach the correct answer. However, to increase the efficiency of the search for negative labels (low utility), Luo et al. (2024b); Dai et al. (2025) implements a binary search algorithm to locate the first step with low utility. One notable variant of step-level utility labels is *advantage*, where the evaluators are not trained to predict the expected reward of each node but the *change* in the expected rewards before and after generating the step (Setlur et al., 2024).

Trace-level utility The coarsest label is the trace-level utility, simply measured by the correctness of the final answer (Lambert et al., 2025).

Both trace-level and step-level utilities do not require human annotation other than the final answer, which is much cheaper to obtain than human annotations (Wang et al., 2024c). However, they cannot serve as a reliable proxy of factuality/coherence/validity due to *unfaithful reasoning*,

Dataset	Train	Eval	Domain	Criteria	# Trace	Human
ROSCOE (Golovneva et al., 2023b)		●	Math, Common	FVU	1.0k	●
RAGTruth [†] (Niu et al., 2024)	●	●	Fact	F	5.9k	●
HaluEval [†] (Li et al., 2023a)	●	●	Fact	F	10k	▲
Math-Shepherd (Wang et al., 2024c)	●		Math	U	440k	×
PRM800k (Lightman et al., 2024)	●	●	Math	V	75k	●
REVEAL (Jacovi et al., 2024)		●	Common	FVC	3.4k	●
MATH-Minos (Gao et al., 2024a)	●		Math	V	440k	×
SCDPO (Lu et al., 2024)	●		Math	U	30k	×
MR-GSM8k (Zeng et al., 2024a)		●	Math	V	3.0k	●
BIG-Bench-Mistake (Tyen et al., 2024)		●	Symbolic	VCU	2.2k	●
CriticBench (Lin et al., 2024)		●	Math, Common, Symbolic	VU	3.8k	×
ProcessBench (Zheng et al., 2024a)		●	Math	V	3.4k	●
MR-Ben (Zeng et al., 2024b)		●	Science, Deductive, Coding	V	6.0k	●
MR-MATH (Xia et al., 2025)		●	Math	VU	0.1k	●
PRMBench (Song et al., 2025)		●	Math	VCU	6.2k	▲
PRM-Clinic (Wang et al., 2025a)		●	Expert(Clinic)	FVC	9.7k	×
VersaPRM (Zeng et al., 2025)	●		Expert	FV	84.1k	×
BiGGenBench [†] (Kim et al., 2025a)		●	Math, Logic	Custom	0.1k	×

Table 3: List of PRM training data and meta-evaluation benchmarks with step-wise annotation. [†] symbol indicates that the datasets include other tasks, such as summarization, instruction following, *etc*, where the # **Trace** column only counts the reasoning subset. **Train/Eval** columns denote if the dataset is used for training or meta-evaluation. **Domain** indicates what tasks are used to sample the reasoning trace. **Criteria** column shows the criteria used to annotate the data classified according to Section 3, where FVCU stands for factuality, validity, coherence, and utility, respectively. BiGGenBench (Kim et al., 2025a) applies query-specific evaluation criteria. Parentheses indicate that the criteria group is not explicitly distinguished in the labels. **Human** column indicates human annotation, where ●▲× denotes full human annotation, automatic annotation/perturbation with human verification, and full LLM-based annotation, respectively.

where traces that reach the correct answer (high utility) often include factual/logical errors (Lanham et al., 2023; Lyu et al., 2023; Zheng et al., 2024a; Kim et al., 2025b).

B.1.2 Inter-annotator agreement

While reasoning trace evaluation is considered more objective than other long-text evaluation tasks (*e.g.*, helpfulness, bias/harmfulness, and language proficiency) (Wang et al., 2024a), a certain amount of inter-annotator disagreement is inevitable. Here, we report the trend in inter-annotator agreement observed in existing human annotation works.

Incorrect solutions for harder problems lead to higher disagreement ProcessBench (Zheng et al., 2024a) consolidates the intuitive hypothesis that inter-annotator disagreement grows when the query is difficult and the trace is incorrect in at least one step. Compared to the easiest case (GSM8k queries, correct trace), where three annotators agree in 95.9% of the cases, the hardest case (OmniMATH, incorrect trace) shows only 47.8% of three-annotator agreement.

Inter-annotator disagreement reflects vagueness in natural language In many cases, the disagreement is significantly affected by the linguistic as-

pects of the reasoning trace. REVEAL (Jacovi et al., 2024) manually classifies steps that aroused disagreement between annotators into 13 distinct categories. Among these, frequent disagreement types like "World knowledge (some world knowledge might not be taken for granted)" and "Unclear reference (one proper noun can refer to multiple real-world entities)" are typical disagreement types observed in simpler *recognizing textual entailment* (natural language inference) tasks (Camburu et al., 2018; Lee et al., 2025), showing that these vagueness is present even in *minimal* settings.

On the other hand, synthetic, algorithmic reasoning tasks like BIG-Bench-Hard (Suzgun et al., 2022) are linguistically uniform. Consequently, BIG-Bench-Mistake that annotate errors in this benchmark (Tyen et al., 2024) observes near-perfect inter-annotator agreement (Krippendorff’s $\alpha > 0.97$), again demonstrating the strong connection between linguistic variation and inter-annotator agreement.

B.2 Label types

Sequence classification The most common label type is sequence classification, where a quality label is assigned to each step/trace. For ex-

ample, Wang et al. (2024c) assigns binary labels to steps based on the utility, and Lightman et al. (2024) assigns ternary validity labels (*correct/incorrect/neutral*) obtained by human annotation. The *neutral* label in Lightman et al. (2024) was introduced to absorb ambiguous cases and minimize inter-annotator disagreement; considering it as positive or negative when training the evaluator does not significantly affect the Best-of-N performance ($<1.0p$).

One caveat of sequence classification is that it is hard to define the labels *after* the first error (propagated error). It is often unclear whether steps that rely on the first erroneous step should be labeled as incorrect (because they rely on incorrect premises) or correct (because the reasoning is correct if assuming the premises are correct) (Jacovi et al., 2024; Mukherjee et al., 2025). Two different label schemas are used to bypass this ambiguity: annotating the *pairwise preference* and annotating the *index of the first erroneous step*.

Preference (win/lose) Reasoning trace evaluation can be formulated as a preference problem (Lai et al., 2024; Lu et al., 2024; Lambert et al., 2025). In this scenario, data points are defined as pairs of reasoning traces, one as the winner and the other as the loser. The pairs are often constructed by sampling two different continuations from a shared prefix or perturbing a correct trace. These data are often used to train the LLM-as-a-value-function models via preference learning algorithms, e.g., DPO (Rafailov et al., 2023).

Identifying first erroneous index Another method is to label the index of the first erroneous step (Zheng et al., 2024a; Zeng et al., 2024a). In this setting, the reasoning trace is given as a list of steps, and the evaluator must predict the index of the first error. If there is no error, the model should predict -1. This setting effectively bypasses the propagated error problem, but converting these labels to binary classification can lead to better performance in sequence classifiers and critic models (Kim et al., 2025b).

C Comparing criteria definitions

C.1 Comparison between proposed definitions

Factuality $\leftrightarrow$ Validity Factuality focuses on the relationship between a step and provided/external knowledge, while validity focuses on the relationship between two model-generated steps. For in-

stance, Given an incorrect step *Albert Einstein died in 1965* (he died in 1955), this step is not factual if the query explicitly mentions that *Einstein died in 1955*. Apart from that, if the previous steps provide the premises for reaching 1955, i.e. *Einstein was born in 1879, and he died at the age of 76*, the step is invalid.

While the standard practice is to treat factuality and validity separately (Prasad et al., 2023; Zhu et al., 2024b; Jacovi et al., 2024), the boundary between stating facts and making logical inferences is often vague, especially in commonsense reasoning. For example, if the step states *Einstein died between 1960 and 1970* when given the information *Einstein died in 1955*, is this step a factual error or a logical error? The boundary heavily relies on the definition of what can be taken as granted, which is also a key factor in defining coherence. REVEAL (Jacovi et al., 2024) delegates the decision to human annotators, and shows that LLMs (Anil et al., 2023; Brown et al., 2020) perform poorly ($F1 < 0.65$) at classifying the steps between factual statements and logical inference.

Validity $\leftrightarrow$ Coherence Existing works often treat coherence as a subtype of validity (Golovneva et al., 2023a; Zhu et al., 2024b; Kim et al., 2025a; Jacovi et al., 2024), as both criteria judge a step based on its previous steps. However, validity and coherence are different by definition, as validity focuses on the logical correctness of a step while coherence focuses on the pragmatic aspect of informativeness. For instance (Figure 3-Coherence), omitting a step (Step 3) from the correct trace will make the subsequent step (Step 3') incoherent, but it is still valid since it can be eventually deduced from the query and previous steps.

Validity $\leftrightarrow$ Utility Previous studies have continuously pointed out that validity does not necessarily lead to utility and vice versa (Lyu et al., 2023; Nguyen et al., 2024). One case is *shortcut reasoning* (Schnitzler et al., 2024; Lee and Hwang, 2025), where LLM generates invalid Chain-of-thoughts but guesses the correct answer directly from the query. ProcessBench (Zheng et al., 2024a) reports that invalid traces with correct answers can be easily found in challenging problems, reaching 51.8% in the olympiad-level Omni-MATH (Gao et al., 2024b).

The distinction between validity and utility has been highlighted by multiple empirical results. Treating these metrics as different yields substantial

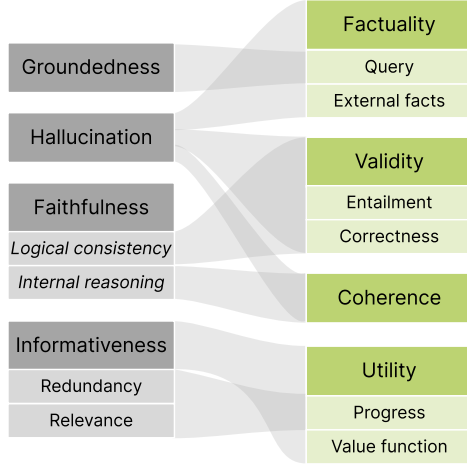


Figure 5: A Sankey diagram displaying the relationship between commonly used terminologies (left) to the proposed taxonomy (right).

performance gain when training sequence classifiers (Zhang et al., 2025) and in Best-of-N decoding (Sun et al., 2024; Kim et al., 2025b). See Section 5 for details.

C.2 Comparison to other definitions

Hallucination is most commonly defined as "*models either generating (1) nonsensical or (2) unfaithful to the source content*" (Ji et al., 2023; Banerjee et al., 2024; Huang et al., 2024b), which corresponds to (1) validity/coherence and (2) factuality. However, some works restrict the meaning of hallucination to factual errors, i.e. "*models generating description tokens that are not supported by the source inputs*" (Xiao and Wang, 2021; Akbar et al., 2024).

Faithfulness is also used with different senses. The most common definition for faithfulness is "*logical consistency between the generated text and the query/previous steps*" (Maynez et al., 2020; Creswell and Shanahan, 2022; Huang et al., 2024b), which includes both factuality (query groundedness) and validity (previous step). Instead, faithfulness can be used as "*accurately representing the model’s internal reasoning process*" (Lyu et al., 2023; Lanham et al., 2023). Under this definition, the final step containing the answer is unfaithful if it is not supported by the previous steps, which falls under the definition of coherence.

Informativeness is defined as "*providing new information that is helpful towards deriving the generated answer*" (Golovneva et al., 2023b; Prasad et al., 2023). Lack of informativeness is often described as **redundancy** "*removing the step does*

not affect the reasoning process" (Chiang and Lee, 2024; Song et al., 2025; Zhou et al., 2024) or **ir-relevance** "*unrelated to the query’s topic or task*" (Wang et al., 2023a; Zhou et al., 2024; Jacovi et al., 2024). Informativeness is synonymous with utility, as it aims to evaluate the contribution of a step to reaching the final answer.

D Details for Discussions

This section provides further details regarding the Discussions (Section 5), specifically Figure 4.

D.1 Estimating Compute

To estimate the compute in Figure 4, we follow the approximation equation from Snell et al. (2024); Kim et al. (2025b). Specifically, the computational cost can be asymptotically approximated as

$$C \in O(N \times L),$$

where C is the total computational cost, N is the number of parameters, and L is the number of tokens. Note that since all compared evaluators use the same base model, N remains constant.

Below, we describe how the computation budget for each method is calculated in Figure 4:

- Unit relative compute (1) corresponds to a single forward pass for an average-length trace. This applies to *Fine-tuned Sequence Classifiers*, as they take the whole trace as the input.
- The *Base Model*, *Fine-tuned Critic Models* (She et al., 2025), and *Fine-tuned LRMs* evaluate each step with a separate forward pass. Thus, the compute is scaled by the number of steps per trace, which is 6.11 on average in ProcessBench (GSM8k + MATH). Note that LRMs like DeepSeek-Distill-Qwen-2.5-7B (DeepSeek-AI, 2025) generate significantly longer traces, with L scaled by 7.58.
- *PARC* (Mukherjee et al., 2025) also uses step-wise critic evaluations, but only using the *Partial context* (average 1.57 premises per step) makes PARC require lower compute by reducing L .
- In *Majority Voting* setting, where 8 step-wise evaluations are sampled per step and aggregated via majority voting, the total computation cost is multiplied by 8.

D.2 Data source

- For *Base model* and *Majority voting* scores, authors conducted experiments with Qwen-2.5-7B-Instruct using the code from [Kim et al. \(2025b\)](#). *Fine-tuned LRM* scores are as reported in the same paper.
- *Sequence classifiers* scores are obtained from [Zhang et al. \(2025\)](#).
- *Partial context* scores are provided by the authors of PARC ([Mukherjee et al., 2025](#)), upon requested by the authors of this survey. While the currently available version of the paper does not contain the result, it will appear in the published version.
- *Fine-tuned PRM* scores are from [She et al. \(2025\)](#).